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・不定積分の接触形 ～累乗型～

α は $\alpha \neq -1$ を満たす実数、 C は積分定数とする。

積分

$$\int \{f(x)\}^{\alpha} f'(x) dx = \frac{1}{\alpha+1} \{f(x)\}^{\alpha+1} + C$$

$$\int e^{f(x)} f'(x) dx = e^{f(x)} + C$$

微分

$$\{f(x)\}^{\alpha}' = \alpha \{f(x)\}^{\alpha-1} f'(x)$$

$$\{e^{f(x)}\}' = e^{f(x)} f'(x)$$

$$\left(\frac{1}{\alpha+1} \{f(x)\}^{\alpha+1} \right)' = \{f(x)\}^{\alpha} f'(x)$$

$$\left(e^{f(x)} \right)' = e^{f(x)} f'(x)$$

(例1) 次の不定積分を求めよ。

C は積分定数とする。

$$\begin{aligned} (1) \int x(x^2+1)^4 dx &= \frac{1}{2} \int (x^2+1)' (x^2+1)^4 dx \\ &= \frac{1}{2} \cdot \frac{1}{5} (x^2+1)^5 + C \\ &= \frac{1}{10} (x^2+1)^5 + C \end{aligned}$$

$$\begin{aligned} (2) \int x^3 \sqrt{2x^2+1} dx &= \frac{1}{4} \int (2x^2+1)' (2x^2+1)^{\frac{1}{2}} dx \\ &= \frac{1}{4} \cdot \frac{3}{4} (2x^2+1)^{\frac{3}{2}} + C \\ &= \frac{3}{16} (2x^2+1)^{\frac{3}{2}} + C \end{aligned}$$

$$\begin{aligned} (3) \int \frac{x^2}{(x^3+1)^3} dx &= \frac{1}{3} \int (x^3+1)' (x^3+1)^{-3} dx \\ &= \frac{1}{3} \cdot \left(-\frac{1}{2}\right) (x^3+1)^{-2} + C \\ &= -\frac{1}{6(x^3+1)^2} + C \end{aligned}$$

$$\begin{aligned} (4) \int \sin x \cos^3 x dx &= -\int (\cos x)' \cos^3 x dx \\ &= -\frac{1}{4} \cos^4 x + C \end{aligned}$$

$$\begin{aligned} (5) \int \frac{\tan x}{\cos^3 x} dx &= \int (\tan x)' \tan x dx \\ &= \frac{1}{2} \tan^2 x + C \end{aligned}$$

$$\begin{aligned} (6) \int \frac{\log x}{x} dx &= \int (\log x)' \log x dx \\ &= \frac{1}{2} (\log x)^2 + C \end{aligned}$$

$$\begin{aligned} (7) \int x e^{-x^2} dx &= -\frac{1}{2} \int (-x^2)' e^{-x^2} dx \\ &= -\frac{1}{2} e^{-x^2} + C \end{aligned}$$

(例2) 次の不定積分を求めよ。

C は積分定数とする。

$$\begin{aligned} (1) \int \cos^3 x dx &= \int \cos x (1 - \sin^2 x) dx \\ &= \int \cos x dx - \int \cos x \sin^2 x dx \\ &= \int \cos x dx - \int (\sin x)' \sin^2 x dx \\ &= \sin x - \frac{1}{3} \sin^3 x + C \end{aligned}$$

$$\begin{aligned} (2) \int \sin^3 x dx &= \int \sin x (1 - \cos^2 x) dx \\ &= \int \sin x dx - \int \sin x \cos^2 x dx \\ &= \int \sin x dx + \int (\cos x)' \cos^2 x dx \\ &= -\cos x + \frac{1}{3} \cos^3 x + C \end{aligned}$$

$$\begin{aligned} (3) \int \tan^4 x dx &= \int \tan^2 x \left(\frac{1}{\cos^2 x} - 1 \right) dx \\ &= \int \frac{\tan^2 x}{\cos^2 x} dx - \int \tan^2 x dx \\ &= \int (\tan x)' \tan x dx - \int \left(\frac{1}{\cos^2 x} - 1 \right) dx \\ &= \frac{1}{3} \tan^3 x - \tan x + x + C \end{aligned}$$

$$\begin{aligned} (4) \int \frac{1}{\tan^3 x \sin^2 x} dx &= -\int \left(\frac{1}{\tan x} \right)' \left(\frac{1}{\tan x} \right)^3 dx \\ &= -\frac{1}{4} \left(\frac{1}{\tan x} \right)^4 + C \\ &= -\frac{1}{4 \tan^4 x} + C \end{aligned}$$

$$\begin{aligned} (5) \int \frac{1}{x(1+\log x)^3} dx &= \int (1+\log x)' (1+\log x)^{-3} dx \\ &= -\frac{1}{2} (1+\log x)^{-2} + C \\ &= -\frac{1}{2(1+\log x)^2} + C \end{aligned}$$